

$$3d^7$$

August 25, 2026

$$1 \quad \left| 3d_{xy}^2 3d_{z^2}^2 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

Here are the CSFs:

$$|1, 3/2, 3/2\rangle = \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|2, 3/2, 3/2\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|3, 3/2, 3/2\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|4, 3/2, 3/2\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|5, 3/2, 3/2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|6, 3/2, 3/2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|7, 3/2, 3/2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|8, 3/2, 3/2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|9, 3/2, 3/2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|10, 3/2, 3/2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle$$

$$\begin{aligned} |11, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\ &+ \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \end{aligned}$$

[illegible]

$$\begin{aligned}
|21, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|22, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|23, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|24, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|25, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|26, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|27, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|28, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|29, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|30, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|31, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|32, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|33, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|34, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|35, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|36, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|37, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|38, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|39, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|40, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|41, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|42, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|43, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|44, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|45, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|46, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|47, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|48, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|49, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|50, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|51, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|52, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|53, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|54, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|55, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|56, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|57, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|58, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|59, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|60, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|61, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|62, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|63, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|64, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|65, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|66, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|67, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|68, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|69, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|70, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|71, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|72, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|73, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|74, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|75, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|76, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|77, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|78, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|79, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|80, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|81, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|82, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|83, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|84, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|85, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|86, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|87, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|88, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|89, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|90, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|91, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|92, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|93, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|94, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|95, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|96, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|97, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|98, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|99, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|100, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|101, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|102, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|103, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|104, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|105, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|106, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|107, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|108, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|109, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|110, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|111, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|112, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|113, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|114, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|115, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|116, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|117, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|118, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|119, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|120, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

The Ground State CSFs are 8 18 28 38

$$\langle 8, \frac{3}{2}, \frac{3}{2} | L_x | 3, \frac{3}{2}, \frac{3}{2} \rangle = -I$$

$$\langle 8, \frac{3}{2}, \frac{3}{2} | L_x | 5, \frac{3}{2}, \frac{3}{2} \rangle = -\sqrt{3}I$$

$$\Delta g_{xx}$$

$$+\frac{2}{3}\zeta\Delta_{8-3}^{-1}$$

$$+2\zeta\Delta_{8-5}^{-1}$$

$$\langle 8, \frac{3}{2}, \frac{3}{2} | L_y | 2, \frac{3}{2}, \frac{3}{2} \rangle = -\sqrt{3}I$$

$$\langle 8, \frac{3}{2}, \frac{3}{2} | L_y | 6, \frac{3}{2}, \frac{3}{2} \rangle = -I$$

$$\Delta g_{yy}$$

$$+2\zeta\Delta_{8-2}^{-1}$$

$$+\frac{2}{3}\zeta\Delta_{8-6}^{-1}$$

$$\langle 8, \frac{3}{2}, \frac{3}{2} | L_z | 10, \frac{3}{2}, \frac{3}{2} \rangle = -2I$$

$$\Delta g_{zz}$$

$$+\frac{8}{3}\zeta\Delta_{8-10}^{-1}$$

For a multiplicity of 4 we have the following:

$$D = \frac{1}{2} (\langle \frac{3}{2}, \frac{3}{2} | \frac{3}{2}, \frac{3}{2} \rangle - \langle \frac{3}{2}, \frac{1}{2} | \frac{3}{2}, \frac{1}{2} \rangle)$$

$$E = \frac{1}{\sqrt{3}} \langle \frac{3}{2}, -\frac{1}{2} | \frac{3}{2}, \frac{3}{2} \rangle$$

$$\left\langle \frac{3}{2}, \frac{3}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle = -\frac{4}{9}\zeta^2 ($$

$$+ \Delta_{10}^{-1} + \frac{1}{4}\Delta_{12}^{-1} + \frac{1}{12}\Delta_{13}^{-1} + \frac{1}{4}\Delta_{15}^{-1} + \frac{1}{12}\Delta_{16}^{-1} + \frac{3}{8}\Delta_{42}^{-1}$$

$$+ \frac{1}{8}\Delta_{43}^{-1} + \frac{3}{8}\Delta_{45}^{-1} + \frac{1}{8}\Delta_{46}^{-1} + \frac{1}{8}\Delta_{62}^{-1} + \frac{1}{24}\Delta_{63}^{-1} + \frac{1}{8}\Delta_{65}^{-1}$$

$$+ \frac{1}{24}\Delta_{66}^{-1} + \frac{1}{4}\Delta_{87}^{-1} + \frac{1}{4}\Delta_{93}^{-1} + \frac{1}{4}\Delta_{96}^{-1} + \frac{1}{4}\Delta_{100}^{-1})$$

$$\begin{aligned}
\left\langle \frac{3}{2}, \frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{1}{2} \right\rangle = & -\frac{4}{9}\zeta^2 (\\
& + \frac{1}{4}\Delta_2^{-1} + \frac{1}{12}\Delta_3^{-1} + \frac{1}{4}\Delta_5^{-1} + \frac{1}{12}\Delta_6^{-1} + \frac{1}{9}\Delta_{20}^{-1} + \frac{1}{3}\Delta_{22}^{-1} \\
& + \frac{1}{9}\Delta_{23}^{-1} + \frac{1}{3}\Delta_{25}^{-1} + \frac{1}{9}\Delta_{26}^{-1} + \frac{1}{8}\Delta_{52}^{-1} + \frac{1}{24}\Delta_{53}^{-1} + \frac{1}{8}\Delta_{55}^{-1} \\
& + \frac{1}{24}\Delta_{56}^{-1} + \frac{8}{9}\Delta_{70}^{-1} + \frac{1}{24}\Delta_{72}^{-1} + \frac{1}{72}\Delta_{73}^{-1} + \frac{1}{24}\Delta_{75}^{-1} + \frac{1}{72}\Delta_{76}^{-1} \\
& + \frac{1}{3}\Delta_{90}^{-1} + \frac{1}{3}\Delta_{97}^{-1} + \frac{1}{12}\Delta_{107}^{-1} + \frac{1}{12}\Delta_{113}^{-1} + \frac{1}{12}\Delta_{116}^{-1} + \frac{1}{12}\Delta_{120}^{-1} \\
&)
\end{aligned}$$

$$\begin{aligned}
\left\langle \frac{3}{2}, -\frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle = & -\frac{4}{9}\zeta^2 (\\
& - \frac{1}{\sqrt{12}}\Delta_{12}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{13}^{-1} + \frac{1}{\sqrt{12}}\Delta_{15}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{16}^{-1} + \frac{\sqrt{3}}{8}\Delta_{42}^{-1} \\
& - \frac{\sqrt{3}}{24}\Delta_{43}^{-1} - \frac{\sqrt{3}}{8}\Delta_{45}^{-1} + \frac{\sqrt{3}}{24}\Delta_{46}^{-1} + \frac{\sqrt{3}}{24}\Delta_{62}^{-1} - \frac{\sqrt{3}}{72}\Delta_{63}^{-1} \\
& - \frac{\sqrt{3}}{24}\Delta_{65}^{-1} + \frac{\sqrt{3}}{72}\Delta_{66}^{-1} + \frac{\sqrt{3}}{12}\Delta_{87}^{-1} - \frac{\sqrt{3}}{12}\Delta_{93}^{-1} - \frac{\sqrt{3}}{12}\Delta_{96}^{-1} \\
& + \frac{\sqrt{3}}{12}\Delta_{100}^{-1})
\end{aligned}$$

$$\begin{aligned}
D = & \frac{2}{9}\zeta^2 (\\
& + \frac{1}{4}\Delta_2^{-1} + \frac{1}{12}\Delta_3^{-1} + \frac{1}{4}\Delta_5^{-1} + \frac{1}{12}\Delta_6^{-1} - \Delta_{10}^{-1} - \frac{1}{4}\Delta_{12}^{-1} \\
& - \frac{1}{12}\Delta_{13}^{-1} - \frac{1}{4}\Delta_{15}^{-1} - \frac{1}{12}\Delta_{16}^{-1} + \frac{1}{9}\Delta_{20}^{-1} + \frac{1}{3}\Delta_{22}^{-1} + \frac{1}{9}\Delta_{23}^{-1} \\
& + \frac{1}{3}\Delta_{25}^{-1} + \frac{1}{9}\Delta_{26}^{-1} - \frac{3}{8}\Delta_{42}^{-1} - \frac{1}{8}\Delta_{43}^{-1} - \frac{3}{8}\Delta_{45}^{-1} - \frac{1}{8}\Delta_{46}^{-1} \\
& + \frac{1}{8}\Delta_{52}^{-1} + \frac{1}{24}\Delta_{53}^{-1} + \frac{1}{8}\Delta_{55}^{-1} + \frac{1}{24}\Delta_{56}^{-1} - \frac{1}{8}\Delta_{62}^{-1} - \frac{1}{24}\Delta_{63}^{-1} \\
& - \frac{1}{8}\Delta_{65}^{-1} - \frac{1}{24}\Delta_{66}^{-1} + \frac{8}{9}\Delta_{70}^{-1} + \frac{1}{24}\Delta_{72}^{-1} + \frac{1}{72}\Delta_{73}^{-1} + \frac{1}{24}\Delta_{75}^{-1} \\
& + \frac{1}{72}\Delta_{76}^{-1} - \frac{1}{4}\Delta_{87}^{-1} + \frac{1}{3}\Delta_{90}^{-1} - \frac{1}{4}\Delta_{93}^{-1} - \frac{1}{4}\Delta_{96}^{-1} + \frac{1}{3}\Delta_{97}^{-1} \\
& - \frac{1}{4}\Delta_{100}^{-1} + \frac{1}{12}\Delta_{107}^{-1} + \frac{1}{12}\Delta_{113}^{-1} + \frac{1}{12}\Delta_{116}^{-1} + \frac{1}{12}\Delta_{120}^{-1})
\end{aligned}$$

$$\begin{aligned}
E = & \frac{4}{9\sqrt{3}}\zeta^2 (\\
& + \frac{1}{\sqrt{12}}\Delta_{12}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{13}^{-1} - \frac{1}{\sqrt{12}}\Delta_{15}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{16}^{-1} - \frac{\sqrt{3}}{8}\Delta_{42}^{-1} \\
& + \frac{\sqrt{3}}{24}\Delta_{43}^{-1} + \frac{\sqrt{3}}{8}\Delta_{45}^{-1} - \frac{\sqrt{3}}{24}\Delta_{46}^{-1} - \frac{\sqrt{3}}{24}\Delta_{62}^{-1} + \frac{\sqrt{3}}{72}\Delta_{63}^{-1} \\
& + \frac{\sqrt{3}}{24}\Delta_{65}^{-1} - \frac{\sqrt{3}}{72}\Delta_{66}^{-1} - \frac{\sqrt{3}}{12}\Delta_{87}^{-1} + \frac{\sqrt{3}}{12}\Delta_{93}^{-1} + \frac{\sqrt{3}}{12}\Delta_{96}^{-1} \\
& - \frac{\sqrt{3}}{12}\Delta_{100}^{-1})
\end{aligned}$$